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**A MINOR PROJECT
PROPOSAL
ON
DECENTRALIZED PREPAID SMART POWER GRID**

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INTRODUCTION

1.1 Background

Electrical utility distribution domains traditionally rely on deeply centralized data hubs and infrastructure networks. In conventional systems, energy consumption data is metered at the consumer's end, transferred across proprietary channels, and recorded on private databases controlled by the central authority. While this model has sustained basic distribution needs for decades, it introduces a systemic vulnerability: asymmetric information control. Consumers have zero real-time architectural visibility into the backend computations, resulting in frequent trust deficits, billing friction, and disputes regarding unexpected or hidden charges. [1]

To address this structural trust gap, this project introduces a decentralized smart grid framework that completely decouples utility transaction computation from central database manipulation. By converging the Internet of Things (IoT), Blockchain architecture, and Machine Learning (ML), the proposed system transitions utility management into a transparent, and self-governing ecosystem.

1.2 Problem Definition

Modern electrical utility structures suffer from a deep trust deficit rooted in asymmetric information control. Because traditional meters stream data into closed, centralized corporate databases, consumers have no way to independently verify the math behind their electricity bills.

On the flip side, utility authorities face massive financial instability due to postpaid billing defaults, where consumers delay payments or completely default on their bills, forcing the utility into expensive and inefficient debt-recovery cycles.

1.3 Objectives

- To eliminate billing opacity, information asymmetry, and utility revenue collection risks by developing a decentralized energy management framework.
- To resolve hidden energy leakages by integrating machine learning models for

predictive budget forecasting and anomaly-based theft detection.

1.4 Scope of Project

The project focuses on providing transparent energy distribution and secure automated prepaid billing from simulated low-voltage IoT edge load sources. It includes real-time analog load sampling, immutable smart contract ledger processing, cryptographic back-end middleware routing, machine learning-based budget forecasting, anomaly-based theft detection, and a dual-interface review dashboard.

LITERATURE REVIEW

2.1 Historical Evolution

Utility metering evolved from electromechanical induction meters (requiring manual reads) to Advanced Metering Infrastructure (AMI) with digital smart meters and wireless telemetry. While AMI automated logging, it preserved centralized, black-box corporate databases. The emerging third generation—decentralized cyber-physical grids—converges IoT, blockchain, and ML to eliminate central middlemen, and executing automated pre-paid billing.

2.2 Algorithms and Models

2.2.1 Smart Contracts (State Machine)

A deterministic Solidity contract dynamically switches between Active (power authorized) and Inactive (credit depleted) states based on real-time hardware telemetry.

2.2.2 Predictive Budgeting (Linear Regression)

A supervised regression framework processes incoming time-series data against historical logs to forecast the exact time of account balance exhaustion.

2.2.3 Anomaly Detection (Isolation Forest)

An unsupervised model maps regular usage profiles and immediately isolates statistical deviations (e.g., sudden voltage drops indicating meter-bypassing).

2.3 Data Processing and Feature Engineering

The system processes two distinct data pipelines: live analog telemetry from the ESP32 edge node and structured synthetic datasets used for training the machine learning models.

Engineered Feature	Data Type	Description and Purpose
Load _{curr}	Float	The real-time simulated power demand scaled to Kilowatts (kW).
Energy _{cum}	Float	The cumulative energy consumed over the session in Watt-hours (Wh).
Delta _{load}	Float	The rate of change in power draw over the last 60 seconds. A sudden, massive drop without a total circuit break is the primary trigger for the theft anomaly detection model.
Hour _{tod}	Integer	A Time-of-Day index (0-23) assigned to the data point to capture cyclical daily consumption habits.
Bal _{token}	Float	The remaining cryptographic token balance pulled via Web3.py from the smart contract, used as the dependent variable for the time-to-exhaustion forecasting model.

2.4 Gaps and Challenge

Existing AMI frameworks rely on vulnerable centralized databases, while pure Web3 frameworks lack physical control systems to enforce ledger states on actual power lines. This project bridges this gap by uniting IoT edge enforcement, ML anomaly detection, and blockchain validation into a single, low-latency automated pipeline.

METHODOLOGY

3.1 Overview

The system samples physical voltage variations, processes them into consumption logs via an API, executes balance deductions on a blockchain, and applies ML to isolate grid anomalies and forecast credit lifespans.

3.2 Proposed Block Diagram

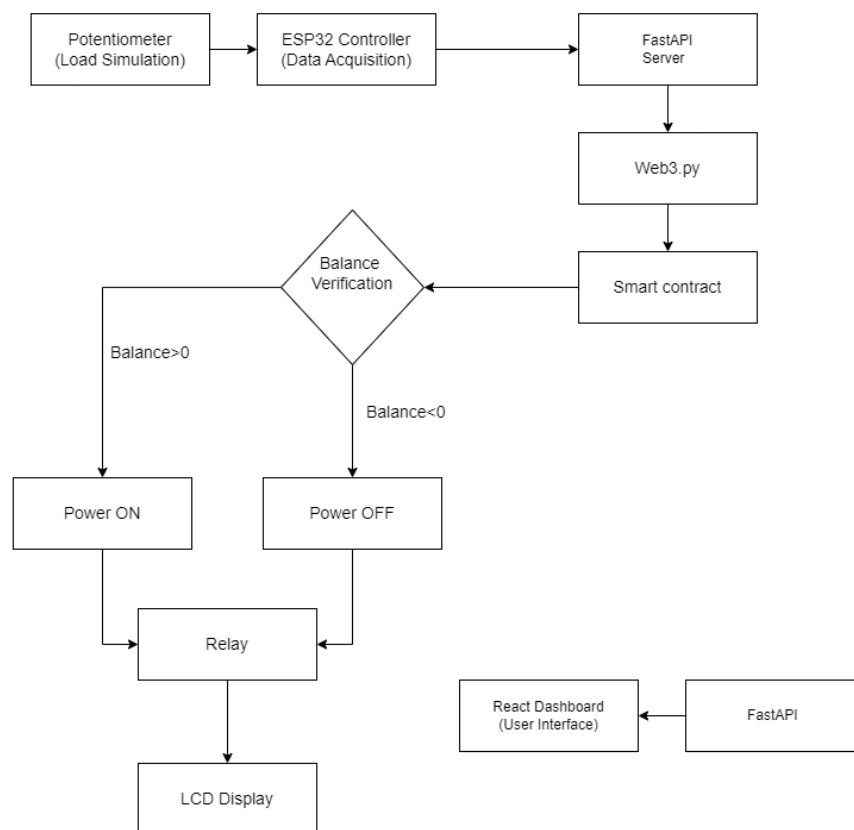


Figure: Proposed Block Diagram of decentralized power grid

3.3 System Design

The architectural framework of the system is structured across three interconnected operational layers that establish a secure, physical-to-digital automation loop.

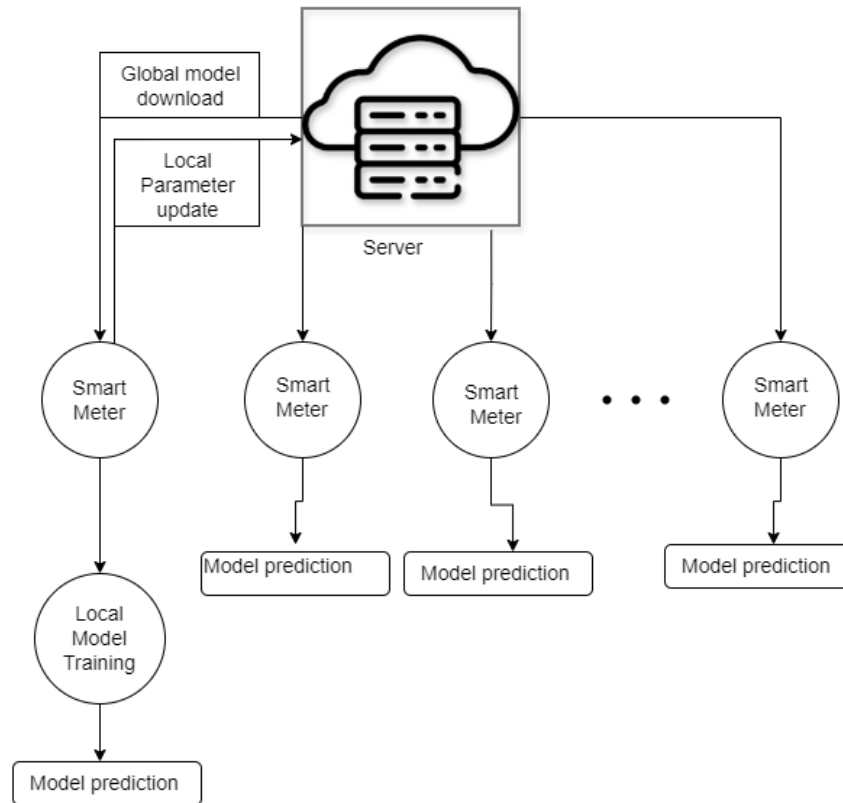


Figure: System Design for decentralized power grid

3.4 Tools and Techniques

- Hardware: ESP32, Potentiometer, Battery
- Software: Arduino IDE (C++), FastAPI (Python), Hardhat (Solidity), React, SQLite

3.5 Data Sources

3.5.1 Primary Sources

Real-time, low-voltage consumption data will be generated during prototype testing by the ESP32 edge node via manual potentiometer manipulation.

3.5.2 Synthetic Data (Machine Learning Training)

To properly train the predictive budgeting and anomaly detection models, a comprehensive synthetic time-series dataset will be programmatically generated using Python. This synthetic dataset will simulate thousands of regular residential consumption cycles based on time-of-day variables.

3.6 Implementation

3.6.1 Hardware Layer Implementation

The physical edge node is built around an ESP32 microcontroller configured within the Arduino IDE using C++. The firmware initializes a continuous hardware execution loop that samples analog signals from the potentiometer via the onboard Analog-to-Digital Converter (ADC) pin.

3.6.2 Blockchain Ledger Implementation

The transaction layer relies on an automated Solidity smart contract compiled, tested, and optimized within a local Hardhat. The contract design establishes state variables that map specific user cryptographic wallet addresses to their remaining token credits and device state logs (Active or Inactive). Prepaid balance tracking is processed immutably using the deduction calculation:

$$\text{Bal}_{\text{new}} = \text{Bal}_{\text{current}} - (\Delta E \times R_{\text{fees}})$$

3.6.3 Backend Middleware and ML Integration

The central logic coordination hub is built using the FastAPI framework in Python. The backend uses the Web3.py library to maintain a persistent connection to the local blockchain environment, listening continuously for emitted ledger events. When telemetry packets arrive from the ESP32, the FastAPI gateway routes the metrics into a Scikit-Learn data pipeline. The machine learning engine concurrently processes the data

through two independent models: a supervised Linear Regression algorithm that evaluates current balance depletion velocities against the inputs to update the countdown time ($T_{\text{remaining}} = \text{Bal}_{\text{new}}/f(\mathbf{X})$), and an unsupervised Isolation Forest algorithm that monitors usage profiles for data anomalies. If a critical anomaly flag or a zero-balance state is confirmed, the backend bypasses standard queues to instantly push an active cutoff signal to the ESP32 firmware while sending an immediate warning alert to the frontend dashboard.

EXPECTED RESULTS

The final prototype will deliver an integrated, low-latency physical automated prepaid grid loop. Operating the linear potentiometer will successfully stream smooth multi-variable JSON consumption telemetry to the FastAPI gateway. The system will achieve complete autonomous enforcement: the exact millisecond the smart contract calculates a zero-token balance, it will emit a ledger event prompting the backend to open the physical ESP32 relay, instantly isolating the consumer's load circuit.

On the software layer, the React-native dashboard will display responsive real-time consumption charts alongside a live, machine-learning-driven countdown timer showing exactly how many hours of power remain. Simultaneously, simulated line-tampering or meter-bypassing events will instantly trigger unsupervised anomaly warning flags on the administrative interface.

The framework's final performance will be quantitatively evaluated using four core engineering metrics:

- Smart Contract Transaction Latency: Time delay between an edge update and ledger balance deduction confirmation.
- Balance Deduction Accuracy: Mathematical precision of contract-executed tariff calculations versus theoretical values.
- Predictive Forecasting Error: The accuracy of the budget exhaustion countdown engine, evaluated using Root Mean Squared Error (RMSE).
- Anomaly Detection Precision: The success rate of the Isolation Forest model in accurately isolating tampering events with minimal false alarms.

EXPECTED SCHEDULE

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